

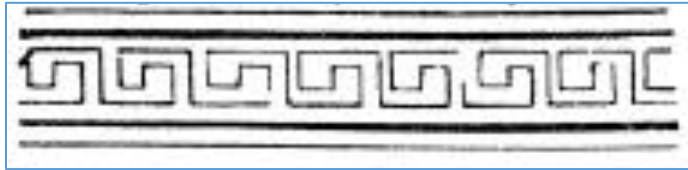
Q1) Symmetry in 1D (30 pts):

Identify the unit cell, lattice, locate all symmetry elements, and identify the line group of the following 10 patterns.

i)



ii)



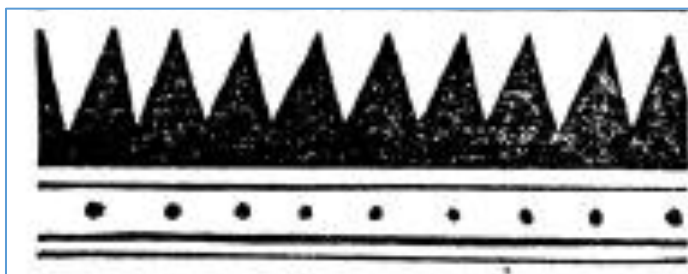
iii)



iv)



v)



vi)



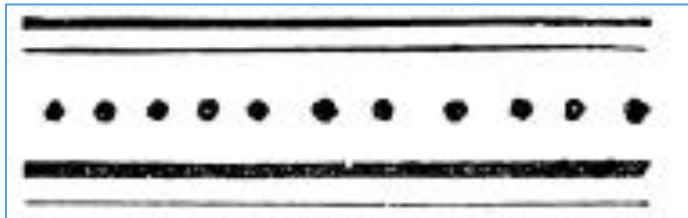
vii)



viii)



ix)



x)

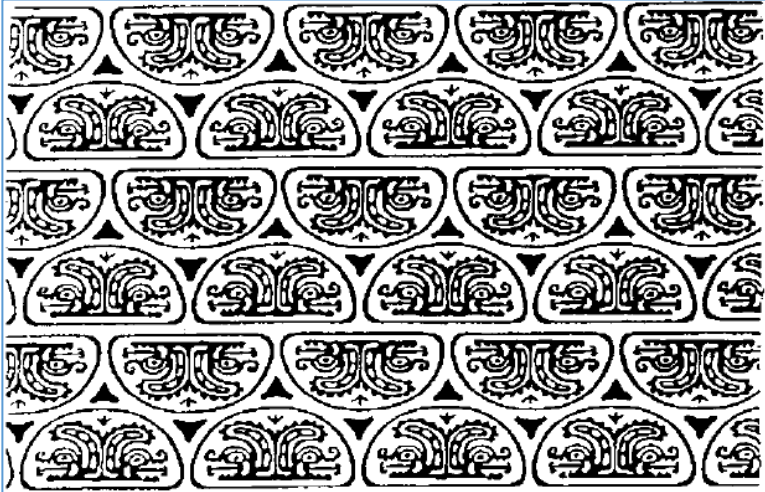


Q2) Symmetry in 2D (20 pts):

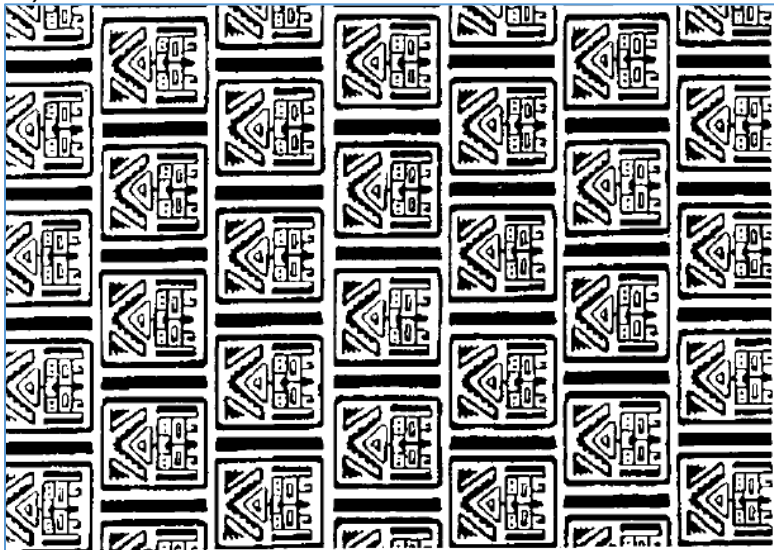
For each of the 2D repeating patterns below:

- Outline a unit cell and identify lattice points in 2D
- Identify and locate all symmetry elements
- Identify the plane group

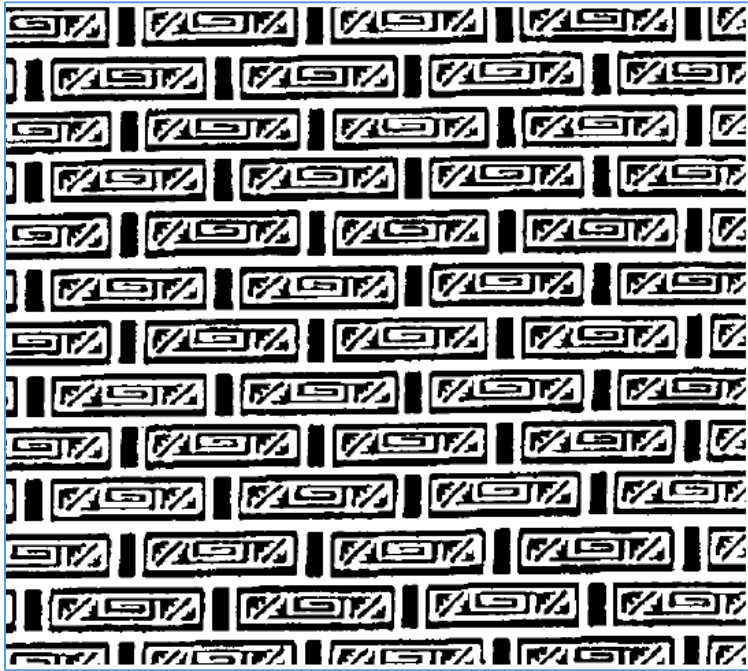
i)



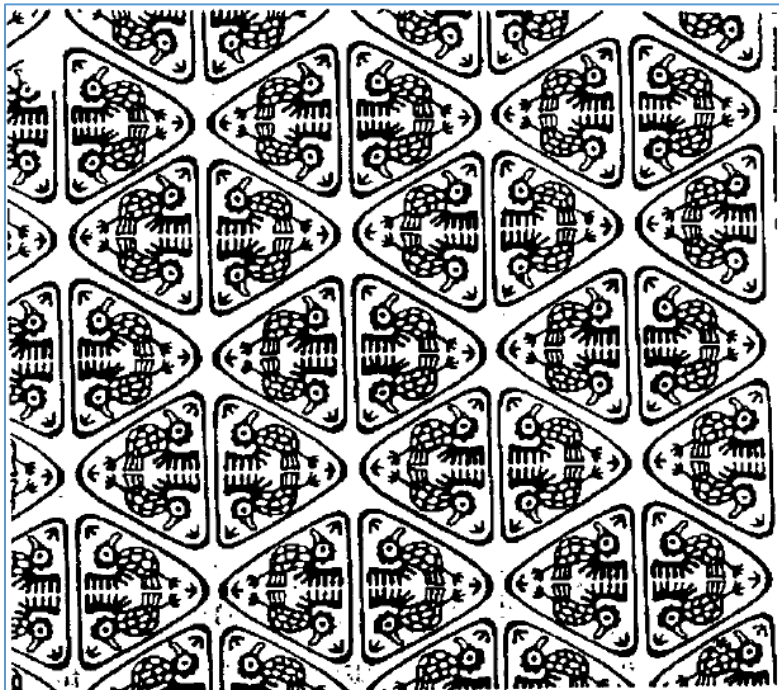
ii)



iii)



iv)



Q3) Decoding Plane Groups (40 pts)

- a) For each plane group below, identify the 2D Bravais lattice and symmetry elements associated to the following plane groups. Are there additional symmetries not explicitly mentioned in the plane group? (18 pts)

p2

cm

pg

p2gg

p4mm

p3

- b) Draw a hexagonal p lattice and illustrate 9 unit cells in 2D. Place a hexad at the corners of the unit cell and assume no other symmetry element is present at the corners. Infer which additional symmetries are present in the 2D crystal. Identify the plane group. (11 pts)
- c) Draw a square lattice and illustrate 9 unit cells in 2D. Place a tetrad at the corners of the unit cell and diagonal mirror lines across the center. Infer which additional symmetries are present within the 2D crystal. Identify the plane group (11 pts)

Q4) Symmetry operations (10 pts + 10 pts bonus)

A) 1D lattice: A point located at $(1/3, 1/6)$ is subject to a glide-reflection line parallel to the y-axis. The Glide-reflection intercepts the x-axis at $x = -1$. Calculate the coordinate of the new point. Assuming these two points form the basis, identify the unit cell, draw symmetry elements, and identify the line group. (10 pts)

B) 2D lattice: A point with coordinates (x, y) is rotated in-plane by a triad located at the origin. Look up the matrix approach (not yet seen in class) and calculate the coordinates of all the points generated by the triad. (**BONUS** 10 pts)